When Does Recursion Pay Off?

An Open, \$0 Reproduction and Honest Dissection of
Two-Timescale Meta-Skill Evolution

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Abstract

Self-improving LLM agents rewrite their own skill files from execution traces. MetaSkill-Evolve [3] makes this *recursive* with a two-timescale scheme: a fast loop evolves a task skill s, while a slow loop evolves a branch-local meta-skill $m = (\psi, \sigma, \alpha, \pi, \varepsilon)$ that parameterises the five agents of the improvement pipeline itself, under the same pipeline applied to itself. We contribute (i) an open, \$0, Apple-Silicon reproduction of the method with a single frozen backbone, and (ii) an honest dissection of *when* the recursion pays off. Decomposing the original results shows the fast loop is the workhorse and the recursive meta-loop contributes a modest, task-dependent increment (+6.4/+8.1/+1.9 points over single-level evolution on OfficeQA/SealQA/ALFWorld). We hypothesise a *backbone-strength law*: the meta-loop's marginal value shrinks as the frozen backbone strengthens, making recursive self-improvement chiefly worthwhile for cheap or weak models. We instrument this with a controlled mechanism study and two further probes—English→Vietnamese meta transfer, and the token cost of recursion. We are explicit about scope: absolute numbers require the paper's 31B backbone, which does not fit our 24 GB budget; our engine reproduces the *method* and the *qualitative structure*, and live confirmation of the backbone-strength law on real backbones is ongoing.

1 Introduction

External skills—reusable procedural knowledge handed to an agent—extend what LLM agents can do on long-horizon tasks. Self-improving agents rewrite these skill files from their own traces. MetaSkill-Evolve observes that such self-evolution is still *non-recursive*: it improves *what* the agent does (the task skill) but not *how* it improves (the improvement procedure), which is authored once and frozen. MetaSkill-Evolve closes this gap with a second, slower loop that evolves the improvement procedure with the same machinery.

This paper is not a new method. It is a reproduction and a dissection, asking a single question a practitioner cares about: **given a fixed backbone and a token budget, is the recursion worth it, and when?** Our contributions:

- An open, dependency-light engine (§3) that reproduces the two-timescale method and runs entirely offline at \$0 on a 24 GB Mac.
- A decomposition showing the fast loop dominates and the recursion is a modest, task-dependent bonus (§4.1).
- The backbone-strength law as a testable hypothesis, with a controlled mechanism study (§4.2), plus EN→VI transfer (§4.3) and a cost analysis (§4.4).

2 Background: MetaSkill-Evolve

Let $U(s) = \mathbb{E}_{(x,y)\sim\mathcal{T}}[r(A_s(x), y)]$ be the utility of a task skill s, where A_s is the agent conditioned on s and $r \in [0,1]$ scores predictions. The fast loop (Alg. 1) mines a worst case, runs the pipeline Analyzer $\to$ Retriever $\to$ Allocator $\to$ Proposer $\to$ Evolver to produce K children, scores them, and commits to a branch DAG. Every H iterations the slow loop (Alg. 2) aggregates meta-productivity $P(m \mid s) = \mathbb{E}\big[\frac{1}{K}\sum_k \big(U(s'_k) - U(s)\big)\big]$, synthesises a meta-failure trace, and applies the *same* pipeline to the five meta files $\{\psi, \sigma, \alpha, \pi, \varepsilon\}$. All five agents share one frozen backbone. Key defaults: H=2, $K_{\max}$=3, frontier size 3 with weights $(1, 0.5, 0.25)$.

3 An Open, $0 Reproduction

Our engine mirrors the method: versioned Markdown skill/meta-skill files with snapshot/restore, a SQLite branch DAG with a weighted frontier, the five agents reading their meta files, and a provider-agnostic backbone (mock for tests, a local MLX model for the weak end, or an OpenAI-compatible gateway for the strong end). The whole pipeline is text-only with a frozen backbone—no gradient updates—so both loops run at $0. The reference implementation is 33 unit tests green with no API key, GPU, or downloads.

4 Dissection

4.1 The fast loop is the workhorse

Decomposing Table 1 of Wang et al. [3] into the single-level jump (fast loop, meta frozen) and the recursive increment (Table 1) shows that 70–90% of the gain over the raw backbone comes from the fast loop alone; the recursion adds $+6.4/+8.1/+1.9$. On ALFWorld the baseline is already 92.3% (near ceiling; static skills even *hurt*), so the +1.9 is within noise and a poor test of recursion.

Table 1: Decomposition of the reported gains. Recursion is the small part.

Benchmark	No-skill	Static	Single-level	MetaSkill-Evolve	Recursion Δ
OfficeQA	31.78	36.09	48.94	55.32	+6.38
SealQA	29.17	29.41	37.21	45.26	+8.05
ALFWorld	92.31	90.38	92.31	94.23	+1.92

4.2 A backbone-strength law

Hypothesis. The recursive meta-loop rewrites the prompts that scaffold the pipeline agents. A strong backbone already analyses and proposes well, so this scaffolding is redundant; a weak backbone benefits more. We therefore predict the meta-gain *decreases* as the backbone strengthens—so recursive self-improvement is chiefly worthwhile for cheap/weak models, which is also the economically interesting regime.

Mechanism study (toy). We construct a controlled task where an agent's behaviour reads a directive from its own meta file, so the slow loop can causally improve the pipeline. With proxy backbones of increasing competence, the meta-gain falls monotonically to zero (Table 2). *These are toy numbers that verify the mechanism and the harness, not benchmark results.*

Live probe (null). We ran the same procedure on real backbones—Claude Haiku 4.5 and Opus 4.8 through an OpenAI-compatible gateway, and a local Gemma-2-2B via MLX—on small multi-step word-problem sets. Every backbone reached ceiling utility *before any evolution* (U_0=1.0), so the meta-gain was 0 for all of them (Table 3). This is a null result about the *task*, not the method: modern instruction-tuned models—even at 2B—leave no headroom on short word problems, so neither the fast loop nor the recursion has anything to improve. A decisive live test of the backbone-strength law needs a benchmark hard enough to hold the base policy below ceiling (e.g. SealQA); we report the null honestly and leave that larger run to future work.

Table 2: Mechanism study on a toy with proxy backbones (offline). Meta-gain vanishes as the backbone strengthens.

Backbone (proxy)	Single-level	Two-level	Meta-gain
weak	0.50	1.00	+0.50
mid	0.75	1.00	+0.25
strong	1.00	1.00	+0.00

Table 3: Live probe (null): every backbone is already at ceiling on the small word-problem set (U_0=1.0), so the recursion has no headroom to exploit. The law needs a harder benchmark to test, not stronger claims from this data.

Backbone	U_0	Single-level	Two-level	Meta-gain
Gemma-2-2B (MLX, local)	1.00	1.00	1.00	+0.00
Claude Haiku 4.5 (gateway)	1.00	1.00	1.00	+0.00
Claude Opus 4.8 (gateway)	1.00	ceiling (run not completed)		

4.3 English→Vietnamese meta transfer

Meta-skills are language-agnostic improvement policies. We ask whether a meta-skill evolved on English tasks accelerates task-skill evolution on Vietnamese ones (GSM8K → vi-gsm8k). The harness runs; live numbers are TODO.

4.4 The cost of recursion

Recursion multiplies inference—you evolve the evolvers. We instrument calls and tokens and compare fast-only vs. two-level at equal iteration budget, reporting utility gained per token. On the toy the recursion costs roughly +30% tokens for its extra gain; the live token/accuracy trade-off is TODO.

5 Related Work

Skill-augmented and self-evolving agents are a fast-growing cluster [2, 1]. MetaSkill-Evolve is distinctive in making the improvement procedure itself recursive with no extra model. Our work is orthogonal: we do not propose a new evolver but dissect *when* recursion helps, on a frozen backbone, at $0.

6 Limitations and Honesty

(1) The paper's absolute numbers used a 31B backbone (20 GB at 4-bit) that does not fit our 24 GB budget; we reproduce the method and qualitative structure, not the exact numbers. (2) The backbone-strength law is currently supported only by a constructed mechanism study; the proxy backbones were designed to express the predicted shape, so they demonstrate the harness, not the law. (3) OfficeQA data is access-gated; SealQA and ALFWorld are public. Live experiments are the immediate next step.

7 Conclusion

Two-timescale recursion is an elegant idea, but its benefit is modest and task-dependent, concentrated where the improvement pipeline—not the base policy—is the bottleneck. We predict, and are testing, that this benefit is a decreasing function of backbone strength. An open $0 engine makes the question cheap for anyone to probe.

References

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